

Correction Notice — Revision 2

*Beyond Sturmian Words in the $3x+1$ Periodicity Conjecture:
A Periodic-Approximation and Arithmetic-Height Bridge*

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Summary. Revision 1 presented the general-intercept case of its main theorem as blocked by one unproved margin lemma, stated there as Conjecture 12.1. That lemma is not open: it is Berthé–Holton–Zamboni, *Initial powers of Sturmian sequences*, Acta Arith. **122** (2006) 315–347, **Propositions 5.1 and 5.2** — Section 5 of the same paper Revision 1 already cites as its primary source. Revision 2 therefore states the main theorem at *every* intercept, and corrects four further defects. The affected material is confined to Revision 1 §§12–13 and its open-problem list; the proved results of §§1–11 are unchanged.

Corrections.

C1. Conjecture 12.1 is a published theorem.

BHZ Prop. 5.1: if $(a_k, a_{k+1}) = (s, t)$ with $s > 1$ for infinitely many k , then *every* Sturmian sequence of that slope has $\text{ice} \geq 2 + \frac{1}{2(s+1)(t+1)+1}$. BHZ Prop. 5.2 covers $(a_k, a_{k+1}, a_{k+2}) = (1, 1, t)$ with $2 + \frac{1}{8t+1}$. A bounded-type slope satisfies one hypothesis or the other by pigeonhole, giving $\text{ice}(\omega) \geq 2 + \frac{1}{2(A_b+1)^2+1}$ at every intercept. The sentences “It is **not** claimed proved here or anywhere in the underlying record” and “until then, that extension remains open” are withdrawn. The weaker claim $\text{ice}(\omega) > 2$ per intercept is likewise already a corollary of BHZ Theorem 1.1.

C2. The “exceptional-orbit reduction” is false.

Revision 1 §12 and its Open 2 relied on $\text{ice}(\omega_\rho) = \text{ind}^*(\gamma)$ off a countable “return” set. BHZ Prop. 2.1(4) gives shift-invariance only off a *finite set of orbits*, and Lemma 2.2 gives equality only *almost everywhere*; a measure-zero set need not be countable. BHZ’s own Prop. 4.1 “keep one” intercept $c_k = a_k - 1$ has $\text{ice} \leq 1 + \theta = 2.618\dots$ at every slope while $\text{ind}^*(\gamma)$ is unbounded, and the admissible family $\{a_k - c_k \in \{1, 2\}\}$ is uncountable with ice uniformly bounded. Consequently intercept 0 is one of the *easiest* intercepts, not the hardest. Open 2 is withdrawn, its premise being false.

C3. Convergent-denominator indexing.

BHZ Prop. 2.7 sets $\gamma = [0; a_1+1, a_2, a_3, \dots]$ with $q_1 = a_1+1$. The record and code used $q_1 = a_1$. Every $\limsup$, hence every ice value reported, is unaffected; every *root length* computed from q_k was wrong. This is the entire explanation of the root-length mismatch the underlying Phase 7 record disclosed against itself; BHZ Prop. 3.3 was never wrong.

C4. Missing length factor.

Revision 1 §13 displayed $S_k \geq A \cdot E_k - C_A D_k - O_A(\log \ell_k)$. The leading term must scale like a length. Correctly, on the odd-weight branch, $S_k \geq \ell_k E_k + 1 - \lceil D(R_k) \rceil \log_2 3 - \log_2(2\ell_k)$, with $\ell_k = |W_k|$ the Sturmian root length. Revision 1’s own flexible condition $E_k \gg \log \ell_k / \ell_k$ is equivalent to $\ell_k E_k \gg \log \ell_k$, so this was a transcription slip and no conclusion drawn from it changes.

C5. Conditional attainment of $y(k)$.

BHZ Prop. 3.3 attains $x(k)$ for every k (root length q_k) but attains $y(k)$ only when $0 < c_k < a_k$

(root length $q_k - c_k q_{k-1}$); Cor. 3.5's three observations are what upgrade the lim sup. Harmless for computing ice, but not for a proof that must supply an *attained* root, since the tightest branch at $A = 1$ is exactly $c_k = a_k$. Revision 2 routes every case through an attained witness explicitly.

Net effect on the results.

- *Strengthened.* The main theorem now reads: for every irrational γ of bounded type, every Sturmian word u of slope γ (any intercept, either mechanical convention), and either complementary symmetric Rote sequence v with $S(v) = u$ (either seed, any shift), $\Phi(v) \notin \mathbb{Q}$.
- *Unchanged.* The abstract irrationality criterion; the discrepancy–height bound; the repetition–height competition; the exact XOR transfer $L_v = L_u + 1$; the density- $\frac{1}{2}$ structural fact; the intercept-0 case and its $\text{term2}(k) > 2 + 1/(A + 1)$ analysis; the counting-route material; the self-correction history; and every numerical finding about structured intercepts — including the identification of the near-maximal-digit family as the empirical worst case, which Revision 2 confirms is *exactly* extremal, with excess exactly $1/(A\beta_A)$, $\beta_A = \frac{A+\sqrt{A^2+4}}{2}$.
- *Superseded or withdrawn.* Conjecture 12.1; Open 1, Open 2, Open 3; the two displayed formulas of C3 and C4.
- *Attribution.* The load-bearing symbolic input is Berthé–Holton–Zamboni's, not this program's. The contribution claimed here is the arithmetic application: the periodic-approximation/arithmetic-height bridge, the exact XOR transfer, the density- $\frac{1}{2}$ fact, the corrected surplus inequality, and the attained-witness bookkeeping with explicit growing root lengths that converts a lim sup bound on ice into an unbounded surplus. Recording a published 2006 theorem as an open conjecture was an error of literature search.
- *Not affected.* No claim is made, in Revision 1 or Revision 2, of progress on the Periodicity Conjecture in general. The live frontier moves to slopes with *unbounded* partial quotients, where BHZ Theorem 1.1 shows some intercept has $\text{ice}(\omega) = 2$ exactly and the odd-weight branch has margin exactly 0.

One gap closed while auditing. The discrepancy step reduces to the rotation by $\gamma/2$, not γ , so it needs $\gamma/2$ to be of bounded type as well — a hypothesis left implicit. It holds: $\|q\gamma\| = \|2q(\gamma/2)\| \leq 2\|q(\gamma/2)\|$, so $\inf_q q\|q(\gamma/2)\| \geq \frac{1}{2} \inf_q q\|q\gamma\| > 0$, and the partial quotients of $\gamma/2$ are bounded by $\lfloor 2(A + 2) \rfloor + 1$. The conclusion $D = O_A(\log \ell)$ is unchanged; its proof is now complete.

Provenance. Revision 2 and this notice were prepared on the isolated branch **phase8-general-intercept** of the **periodicity-conjecture-bridge** repository. The Phase 1–7 research record was not rewritten: the Phase 7 files are preserved verbatim and carry conspicuous correction banners pointing to **theorems/phase7/PHASE7_CORRECTIONS.md**. Reproducible diagnostics are under **scripts/phase8/** with outputs in **data/phase8/**. Revision 2 is intended as a *new Zenodo version* of the existing record, leaving Revision 1 retrievable.